

PARTS LIST

Semiconductors:

IC1	- LM741 op-amp
IC2, IC3	- NE555 timer
IC4	- 4017B decade counter
IC5	- 4027 dual J-K master/slave flip-flop
LED1-LED6	- 5mm LED
T1, T3, T4	- BC548 npn transistor
T2	- BC538 pnp transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R4, R8, R9,	
R12-R15	- 10-kilo-ohm
R5	- 150-kilo-ohm
R6	- 100-kilo-ohm
R7	- 22-kilo-ohm
R10	- 22-ohm
R11	- 68-kilo-ohm
R16-R21	- 1-kilo-ohm
VR1	- 100-kilo-ohm preset

Capacitors:

C1, C2	- 100nF ceramic disk
C3, C5	- 10 μ F, 16V electrolytic
C4, C6, C7	- 10nF ceramic disk
C8, C9	- 100 μ F, 16V electrolytic

Miscellaneous:

CON1, CON2	- Two-pin connector
MIC	- Electret microphone

by transistor T2, which rapidly discharges the timing capacitor C5 if a pulse arrives when the timer is already running. This means that once triggered, the output of IC3 will remain high as long as additional trigger pulses are received within its set timing period.

Pulses from the collector of T1 are also applied to the clock input (pin 14) of IC4. Initially, output '0' (pin 3) of the counter is high, illuminating LED1. The first pulse advances the count to light up LED2, indicating that the circuit is active and ready to receive a handclap command.

Every time a pulse (clap) is received before the timing period of IC3 elapses, the process repeats, incrementing the counter by one. When the timing period of IC3 elapses with no claps detected for approximately 750 ms, the output (pin 3) of IC3 goes low, turning transistors T3 and T4 off.

The rising voltage on transistor T3 clocks the two J-K flip flops in IC5. Input to IC5 comes from outputs O2 through O5 of IC4. Devices can be connected through a relay driven by the output of IC5. Here we have connected only LEDs to represent the devices.

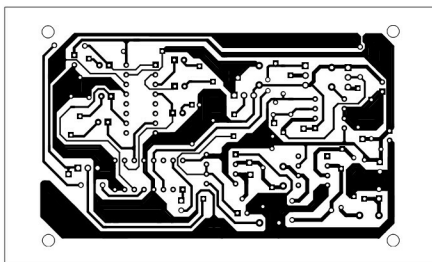


Fig. 2: An actual-size, single-side PCB for the clap-controlled switch

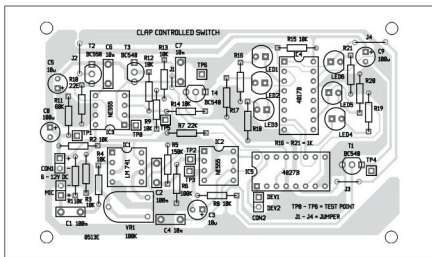


Fig. 3: Component layout for the PCB

Considering all the possible logic states of a J-K flip flop, the commands will therefore be:

1. Two claps: Turn Device 1 'on'
2. Three claps: Turn Device 1 'off'
3. Four claps: Turn Device 2 'on'
4. Five claps: Turn Device 2 'off'

Finally, the rising voltage on the collector of T4 resets the counter, making it ready for the next clap sequence. Note that the additional 10nF capacitors between the reset input (pin 15) and ground in conjunction with resistor R15 add a short delay to the reset signal. This ensures that the counter is not reset until the J-K flip flops have been clocked. If desired, you can easily expand the circuit by adding more flip-flops and counters.

Construction and testing

An actual-size, single-side PCB for the clap-controlled switch is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB to save time and minimise assembly errors. Carefully assemble the components and double-check for any overlooked error. Connect devices via the basic relay circuit using triggering inputs from CON2.

To test the circuit, check the supply voltage at TP1 with respect to TP0. Now clap once and check voltage levels at all the test points using an oscilloscope as mentioned in the table. ●

The author is a regular contributor to EBY and has many articles published to his credit in India and abroad